Arunabha Majumder

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Portfolio: arunabha22.github.io/arunabha-majumder

Medtronic, Surgical Robotics, Galway

Principal Systems Engineer, Robotics Capital & Instrument Performance (R81151)

Dear Medtronic hiring team,

I'd like to apply for the Principal Systems Engineer role in Robotics Capital and Instrument Performance. Taking what a person needs from a robot, turning it into numbers I can test, and then proving it on real hardware is the core of my work, and doing that for surgical robots that help patients is where I want to go.

In my research at Aalborg University I designed a hybrid shoulder exoskeleton. I started from the torque a person needs to lift the arm and split it between two subsystems: a variable stiffness mechanism, with a worm gear that pretensions the spring, and a 6 Nm motor working in parallel. I optimised the design with a genetic algorithm (diameter, spring stiffness and pretension force), integrated the motor, encoder and force/torque sensors, and verified the stiffness the system can provide on a physical test setup. It uses over 25% less motor energy and won the Best Paper Award at ISRM 2026.

I also designed and modelled a novel 5-bar planar parallel robot with variable stiffness mechanisms (VSM), estimated its stiffness with less than 10% error and quantified human arm impedance. Force came from the variable stiffness actuators and the robot's Jacobian, without a force sensor at the end effector, and I checked the model against measured results. For the arm impedance study with this robot (6 participants), I implemented an Arduino-based trigger so the robot's motion and the Qualisys recording stayed in sync, and I collected EMG data from 12 participants in another study.

In my Master's at CSIR-CMERI I worked in a team of 4 electronics and mechanical engineers on a soft exosuit, and designed the outer PID loop, with integral anti-windup and error nonlinearisation, of a neural network-based gain-scheduled controller in MATLAB/Simulink and LabVIEW (0.3 to 0.78% error). I have taught university courses and supervised Bachelor's students. I also hold a Senior Diploma in Fine Arts and do graphic design (Illustrator, Photoshop, Procreate), which helps me explain complex systems clearly to different teams.

I would love to bring new and innovative ideas to your surgical robotics team and grow into systems leadership with you. My portfolio, with videos and results from my work, is at arunabha22.github.io/arunabha-majumder. Thank you for reading.

Best,
Arunabha Majumder